

Lecture 15 : Routing examples

Introduction

In the vast and interconnected landscape of the internet, billions of data packets constantly traverse complex networks, each seeking its specific destination. This intricate dance of data relies heavily on sophisticated routing algorithms and protocols that guide IP packets from their source to their intended recipient through a series of intermediate routers. Understanding how these routers make forwarding decisions is fundamental to comprehending network operations. This lecture delves into practical examples of how routers utilize their routing tables to process incoming IP packets, determine the correct outgoing interface, and ensure efficient data delivery. By examining various routing configurations and packet forwarding scenarios, we will gain insight into the core mechanics that underpin all network communication, laying the groundwork for understanding both robust network design and potential vulnerabilities.

Core Concepts

Term	Definition	Significance
Routing Table	A data structure within a router that stores information about network destinations and the paths to reach them.	Essential for a router to determine where to send incoming packets.
Subnet Mask	A 32-bit number that distinguishes the network portion of an IP address from the host portion.	Crucial for identifying the network segment to which a destination IP address belongs.
Bit-by-bit ANDing	A logical operation performed between an IP address and a subnet mask to extract the network address.	The fundamental step in matching a packet's destination to a routing table entry.
Longest Prefix Match	A rule stating that if multiple routing table entries match a packet's destination IP address, the	Ensures the most specific and often most efficient route is selected when ambiguity exists.

Term	Definition	Significance
	entry with the longest matching network prefix (most bits) is chosen.	
Default Route	A special routing table entry (often 0.0.0.0/0) that specifies where to send packets for which no other specific route is found.	Provides a fallback mechanism, ensuring packets are not dropped if no explicit match exists.
CIDR Notation	Classless Inter-Domain Routing notation, a compact way to represent IP addresses and their associated subnet masks (e.g., 192.168.1.0/24).	Simplifies the representation of network addresses and mask lengths, especially in modern routing tables.
Gateway/ Next Hop	The IP address of the next router in the path to a destination network, used when the destination is not directly connected.	Directs packets through intermediate routers to reach distant networks.
Interface	A physical or logical connection point on a router through which packets are sent or received.	Specifies the physical outgoing link a packet must traverse to reach its next destination.

Detailed Analysis

1. Basic Packet Forwarding with Subnet Masks

The fundamental process of packet forwarding involves a router receiving an IP packet and consulting its routing table to determine the appropriate outgoing interface. This decision is made by comparing the packet's destination IP address with entries in the routing table.

- **The Matching Process:** For each entry in the routing table, the router performs a bit-by-bit logical AND operation between the packet's destination IP address and the entry's `subnet mask`. The result of this ANDing operation is then compared to the `destination network address` specified in that routing table entry.

- **Example Scenario:** Consider a router with four interfaces (eth0, eth1, eth2, eth3) and a routing table containing destination network addresses, subnet masks, and corresponding interfaces.
- An IP packet arrives with the destination address `128.35.57.16`.
- **Row 1 Check:** The first entry has a subnet mask of `255.255.255.0`.
- ANDing `128.35.57.16` with `255.255.255.0` yields `128.35.57.0`.
- This matches the destination network address `128.35.57.0` in the first row.
- **Row 2 Check:** The second entry has a subnet mask of `255.255.255.128`.
- ANDing `128.35.57.16` with `255.255.255.128` yields `128.35.57.0` (since 16 AND 128 in binary results in 0).
- This also matches the destination network address `128.35.57.0` in the second row.
- **Multiple Matches:** In this case, both the first and second rows provide a match. This is where the `longest prefix match` rule becomes critical.

2. Resolving Ambiguity with Longest Prefix Match

When multiple entries in a routing table match a packet's destination, the router must choose the most specific route. This is achieved using the `longest prefix match` rule.

- **The Rule:** The router selects the routing table entry where the `subnet mask` specifies the largest number of bits for the network address. A longer prefix indicates a more specific route.
- **Applying to Example 1:**
 - For the first matching row, the subnet mask `255.255.255.0` corresponds to 24 network bits (8+8+8).
 - For the second matching row, the subnet mask `255.255.255.128` corresponds to 25 network bits (8+8+8+1, as 128 is `10000000` in binary).
 - Since 25 bits is a longer prefix than 24 bits, the second entry is chosen.
 - The packet `128.35.57.16` will be forwarded to `eth1`.
- **Handling No Match:** If no entry in the routing table matches the destination IP address, the packet is forwarded via the `default route`. This is typically an entry with a destination address of `0.0.0.0` and a subnet mask of `0.0.0.0` (or `/0` in CIDR notation), often pointing to an interface connected to an external router. For instance, if a packet with destination `192.112.17.25` doesn't match any specific entry, it would be sent to `eth3` if that's the `default route`.

3. Subnetting and Interface-Specific Forwarding

Routers often connect to multiple subnets, each requiring specific forwarding rules.

- **Scenario:** A router is connected to four subnets (a, b, c, d) via interfaces a, b, c, d, respectively, and an external router via interface e. All subnets share a common prefix `215.1.2.x`, with the last octet defining the subnet. The subnet mask for these subnets is `255.255.255.192` (meaning the first two bits of the last octet are part of the network address).
- Subnet a: `215.1.2.0` (range 0-63)
- Subnet b: `215.1.2.64` (range 64-127)
- Subnet c: `215.1.2.128` (range 128-191)
- Subnet d: `215.1.2.192` (range 192-255)
- **Packet Examples:**
 - **Packet 1: `215.1.2.33` :**
 - ANDing `33` (binary `00100001`) with `192` (binary `11000000`) yields `0` (binary `00000000`).
 - This matches `215.1.2.0`, so the packet is forwarded to interface `a`.
 - **Packet 2: `215.1.2.78` :**
 - ANDing `78` (binary `01001110`) with `192` (binary `11000000`) yields `64` (binary `01000000`).
 - This matches `215.1.2.64`, so the packet is forwarded to interface `b`.
 - **Packet 3: `215.1.2.144` :**
 - ANDing `144` (binary `10010000`) with `192` (binary `11000000`) yields `128` (binary `10000000`).
 - This matches `215.1.2.128`, so the packet is forwarded to interface `c`.
 - **Packet 4: `215.1.2.200` :**
 - ANDing `200` (binary `11001000`) with `192` (binary `11000000`) yields `192` (binary `11000000`).
 - This matches `215.1.2.192`, so the packet is forwarded to interface `d`.

4. Advanced Routing Tables with CIDR and Gateway Information

Modern routing tables often include more details, such as `CIDR notation` for subnet masks and `gateway/next hop` addresses for indirect routes.

- **CIDR Notation:** Instead of a dotted-decimal subnet mask, a `/` followed by the number of network bits is used (e.g., `/22` means 22 network bits).

- **Gateway/Next Hop:** If a destination is not directly connected to the router, the routing table specifies the IP address of the next router (gateway) to which the packet should be sent. A flag (e.g., 'G') might indicate if it's a gateway route (indirect) or a direct route.
- **Example Packet 135.46.63.10 :**
 - The routing table has entries like 135.46.56.0/22 and 135.46.60.0/22 .
 - For 135.46.63.10 with a /22 mask:
 - The first two octets (135.46) are fixed.
 - The next 6 bits of the third octet are part of the network. 63 in binary is 00111111 . Taking the first 6 bits gives 001111 , which is 60 in decimal.
 - So, 135.46.63.10 with a /22 mask resolves to 135.46.60.0 .
 - This matches the entry 135.46.60.0/22 , and the packet is forwarded to the specified interface/gateway.
- **Default Route in Practice:** A default route is typically represented as 0.0.0.0/0 . If a packet's destination (135.46.52.2 in an example) does not match any specific entry, it will be forwarded via the default route to an external gateway.

5. Optimizing Routing Table Lookups

The efficiency of routing table lookups is crucial for router performance.

- **Sorting for Efficiency:** To speed up the longest prefix match process, routing tables can be sorted.
- If entries are sorted in descending order of their subnet mask length (i.e., longest prefixes first), the first match found during a lookup will automatically be the longest prefix match . This eliminates the need to check all other entries once a match is found, saving search and matching time.
- **Example of a Constructed Routing Table:** For a router R1 connected to networks 180.70.64.0/26 , 180.70.65.0/26 , 201.4.16.0/22 , 201.4.22.0/24 , and a default route , the routing table would list these networks with their respective interfaces (e.g., m0, m1, m2, m3) and the default route.
- A packet with destination 180.70.65.140 would match 180.70.65.0/26 and be forwarded via m0 .
- A packet with destination 201.4.22.35 would match 201.4.22.0/24 (as it's a longer prefix than /22) and be forwarded via m3 .

Key Takeaways

- [] IP packet forwarding relies on routers consulting their routing tables.
- [] The core matching process involves bit-by-bit ANDing of the destination IP with the subnet mask.
- [] The `longest prefix match` rule is essential for resolving multiple matching routing table entries.
- [] If no specific route matches, packets are forwarded via the `default route` (0.0.0.0/0).
- [] `CIDR notation` provides a concise way to represent network addresses and mask lengths.
- [] `Gateway/next hop` addresses are used for indirect routes to non-directly connected networks.
- [] Sorting routing table entries by mask length (longest first) optimizes the `longest prefix match` lookup process.
- [] Understanding routing mechanisms is critical for network design, troubleshooting, and cybersecurity analysis.

Conclusion

The examples explored in this lecture demonstrate the precise, logical steps a router takes to forward an IP packet. From the basic bit-by-bit ANDing with a subnet mask to the sophisticated application of the `longest prefix match` rule, every decision is a calculated step towards the packet's destination. The concept of a `default route` ensures network resilience, preventing packets from being dropped even when specific routes are unknown. Furthermore, the use of `CIDR notation` and `gateway/next hop` addresses reflects the practical complexities of modern internet routing.

Beyond efficient data delivery, a deep understanding of routing is paramount in the field of cybersecurity. As discussed, routing tables are the very fabric of network traffic control. If a malicious actor gains control of a router and modifies its routing table, packets can be arbitrarily redirected, intercepted, or dropped, leading to severe network disruptions, data breaches, or denial-of-service attacks. Therefore, mastering the principles of routing is not just about ensuring connectivity; it's about safeguarding the integrity and security of our digital infrastructure. These foundational concepts will prove invaluable in later discussions on network-based attacks and defensive strategies in ethical hacking.